Inverse matrices and inverse maps

All matrices and vectors are real. Let A be an $n \times n$ matrix.

Invertible / nonsingular: there is an $n \times n$ matrix B such that $AB = BA = I_n$. The inverse is unique and is denoted by A^{-1}. If no inverse exists, A is **singular / noninvertible**.

One-sided test for square matrices: for $n \times n$ matrices A, B, $AB = I_n$ implies $BA = I_n$, so $B = A^{-1}$.

Inverse identities: if $A, B, A_1, \dots, A_k$ are invertible $n \times n$ matrices, then their products, inverses, and transposes are invertible, with

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1},$$

$$(A^{-1})^{-1} = A, \qquad (A^T)^{-1} = (A^{-1})^T.$$

Inverse transformation: $T : \mathbb{R}^n \to \mathbb{R}^n$ has an inverse if a map $S : \mathbb{R}^n \to \mathbb{R}^n$ satisfies $S(T(\mathbf{x})) = T(S(\mathbf{x})) = \mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^n$. For $T_A(\mathbf{x}) = A\mathbf{x}$, an inverse exists exactly when A is invertible; it is unique and $(T_A)^{-1} = T_{A^{-1}}$.

Solving a system: if A is invertible, then for every $\mathbf{b} \in \mathbb{R}^n$, the unique solution of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b}$. In particular, $A\mathbf{x} = \mathbf{0}$ has only the solution $\mathbf{x} = \mathbf{0}$.

Equivalent invertibility criteria

For an $n \times n$ matrix A, the following conditions are equivalent:

1. A is invertible.
2. The columns of A are linearly independent.
3. The columns of A span $\mathbb{R}^n$.
4. The rows of A are linearly independent.
5. The rows of A span $\mathbb{R}^n$.
6. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
7. A is row equivalent to I_n.
8. $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^n$.
9. $\operatorname{rank}(A) = n$.

The two-by-two determinant and inverse

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad \det(A) = ad - bc.$$

A is invertible if and only if $ad - bc \neq 0$. In that case,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Minors, cofactors, and general determinants

Let $A = [a_{ij}]$ be $n \times n$, with $n \geq 3$ and $1 \leq i, j \leq n$.

Minor: $\text{minor}(A)_{ij}$ is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A. For $n = 3$, each minor is a two-by-two determinant.

Cofactor: attach the sign determined by the original row and column:

$$\text{cof}(A)_{ij} = (-1)^{i+j} \, \text{minor}(A)_{ij}.$$

The sign is positive when $i + j$ is even and negative when $i + j$ is odd.

Determinant: define recursively by cofactor expansion along the first row:

$$\det(A) = \sum_{j=1}^{n} a_{1j} \, \text{cof}(A)_{1j}.$$

Start with the two-by-two formula; this defines the three-by-three determinant, then each larger size using determinants of the preceding size.

Expansion along any row or column: every choice gives the same determinant. For any fixed row i, and for any fixed column j, respectively,

$$\det(A) = \sum_{j=1}^{n} a_{ij} \, \text{cof}(A)_{ij}, \qquad \det(A) = \sum_{i=1}^{n} a_{ij} \, \text{cof}(A)_{ij}.$$

Each term is an entry times its own cofactor, including the cofactor's sign.

Triangular and diagonal matrices

Let $A = [a_{ij}]$ be $n \times n$.

Upper triangular: $a_{ij} = 0$ for $i > j$ (below the main diagonal).

Lower triangular: $a_{ij} = 0$ for $i < j$ (above the main diagonal).

Diagonal: both upper and lower triangular; $a_{ij} = 0$ whenever $i \neq j$. Write $\text{diag}(\lambda_1, \ldots, \lambda_n)$ for the diagonal matrix whose main diagonal is $\lambda_1, \ldots, \lambda_n$.

Triangular determinant: if A is upper or lower triangular, $\det(A) = a_{11} \cdots a_{nn}$. In particular, $\det(I_n) = 1$.

Determinant identities and invertibility

For $n \times n$ matrices A, B:

Zero row or column: if any row or column of A is entirely zero, then $\det(A) = 0$.

Product: $\det(AB) = \det(A) \det(B)$.

Transpose: $\det(A^T) = \det(A)$.

Invertibility test: A is invertible if and only if $\det(A) \neq 0$. When A is invertible, $\det(A^{-1}) = 1/\det(A)$.

Orthogonal matrices

Orthogonal matrix: a square matrix A with $A^T = A^{-1}$. Equivalently, $A^T A = AA^T = I_n$; the transpose is its inverse.